

Data Science Program, George Washington University

Sample Midterm Exam

DATS 6101 (Introduction to Data Science)

Date: **October 26, 2026**

Instructor: Prof. Sushovan Majhi

Name: _____

Please read the following instructions carefully.

- This is a **sample paper** for practice only, at the real length and the real difficulty. It contains 13 questions for a total of 100 points, covering everything through Oct 19.
- You have **2 hours 30 minutes**, the full session. Answer all questions.
- The exam is **open-book** and **open-note**, but **printed material only**: the chapters covered, the labs whose solutions have been released, the homework covered, and your own notes.
- **Laptops, tablets, phones** and **watches** must be **switched off and put away**. No AI assistance and no communication of any kind.
- Nothing has to be computed that cannot be done on paper: every critical value a question needs is **printed in the question**. A correct number with no sentence around it earns part marks.
- All notation has its usual meaning. You may use standard results directly by stating them appropriately. **Read the instructions for individual questions carefully** before answering.

Question	Points	Score
1	20	
2	8	
3	6	
4	6	
5	6	
6	7	
7	7	
8	8	
9	6	
10	6	
11	7	
12	8	
13	5	
Total:	100	

Part A Reading a claim

1. For each statement, say whether it is **defensible as written**, and give **one sentence** saying why. A verdict with no sentence earns one point of the four.

(a) (4 points) “The 95% confidence interval for mean session length is 4.2 to 5.8 minutes, so 95% of sessions last between 4.2 and 5.8 minutes.”

(b) (4 points) “We got $p = 0.03$, so there is a 3% probability that the null hypothesis is true.”

(c) (4 points) “The difference was not significant ($p = 0.41$), so the two page designs perform the same.”

(d) (4 points) “Our sample is 50,000 responses, so the estimate is accurate to within a fraction of a percent.”

(e) (4 points) “ $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.”

SAMPLE

Part B Probability and random variables

2. A screening test for a condition present in **2%** of a population detects it in **95%** of the people who have it, and is negative in **90%** of the people who do not.

(a) (6 points) A randomly selected person tests positive. What is the probability they have the condition? Show the working.

(b) (2 points) In one sentence, say what makes the answer so much smaller than 95%.

3. (6 points) A support contract pays out per incident as follows.

Payout	\$0	\$50	\$200	\$500
Probability	0.70	0.20	0.07	0.03

Compute $E[X]$ and $\text{Var}(X)$, showing the working.

4. Of 200 visitors, 120 arrived on mobile and 80 on desktop. 30 of the mobile visitors converted, and 20 of the desktop visitors converted.

- (a) (4 points) Are “converted” and “arrived on mobile” independent for these visitors? Show the check.
- (b) (2 points) A colleague concludes the device does not *cause* a difference in conversion. In one sentence: do the data support that?

Part C Expectation, variance and the CLT

5. X has mean 12 and variance 9.

(a) (2 points) Give $E[3X - 2]$ and $\text{Var}(3X - 2)$.

(b) (2 points) Give $E[\bar{X}]$ and $\text{Var}(\bar{X})$ for the mean of 36 independent draws.

(c) (2 points) Give the standard error of that mean.

6. A survey collects 1,200 responses, 30 from each of 40 offices. People in the same office answer alike. A simulation of the design shows the true standard error of the mean is **three times** what $s/\sqrt{1200}$ reports.

(a) (4 points) How many *independently sampled* people would give the same standard error? Show the reasoning.

(b) (3 points) Which step of the derivation of $\text{SE}(\bar{X}) = \sigma/\sqrt{n}$ fails here, and in which direction does software get the answer wrong?

7.

- (a) (4 points) Which of these does the central limit theorem license? Mark each yes or no.
- (i) The sampling distribution of $\bar{X}$ is approximately Normal for large n .
 - (ii) The data become approximately Normal for large n .
 - (iii) The sampling distribution of the sample median is approximately Normal.
 - (iv) It applies whatever the shape of the population, provided the variance is finite and the draws are independent.
- (b) (3 points) A colleague says “ n is over 30, so we are fine.” Give the better answer, and say what you would do instead.

Part D Confidence intervals

8. A sample of **100** support tickets has mean resolution time $\bar{x} = 52.0$ minutes and standard deviation $s = 20.0$ minutes. Use $t_{0.975, 99} = 1.984$.

(a) (4 points) Compute the standard error, then the 95% confidence interval. Show the working.

(b) (2 points) Write the one sentence you would put in the report.

(c) (2 points) A reader asks whether 95% of tickets are resolved inside that interval. Answer in one sentence.

9. A pilot converts **6 of 200** visitors. The Wald interval is $\hat{p} \pm 1.96\sqrt{\hat{p}(1 - \hat{p})/n}$, and here $\sqrt{\hat{p}(1 - \hat{p})/n} = 0.0121$.

(a) (2 points) Compute the interval.

(b) (4 points) Give two distinct reasons not to report it, and say what you would report instead.

10.

- (a) (2 points) A 95% interval is 8 units wide at $n = 400$. How many observations would make it 4 units wide, all else equal?
- (b) (2 points) Moving from 95% to 99% confidence does what to the width, and why?
- (c) (2 points) An analyst narrows an interval by dropping the six largest values. In one sentence, what have they actually reported?

Part E Testing

11. A team changes the checkout flow and wants to know whether mean basket value moved. They have 900 sessions on the old flow and 900 on the new.

(a) (2 points) State H_0 and H_1 .

(b) (2 points) Name the test.

(c) (3 points) They get $p = 0.002$. Say precisely what that licenses, and name one thing it does **not** tell them.

12. For each scenario, name the test you would run and give one line of justification.

(a) (2 points) Forty students take a diagnostic before the course and the same diagnostic after; you want to know whether scores moved.

(b) (2 points) A sample of 800 users is asked yes or no; you want an interval for the share saying yes.

(c) (2 points) Two independent groups of 60, each rating satisfaction 1–10; you want to know whether the average rating differs.

- (d) (2 points) A sample of 1,000 sessions, and you want to know whether the conversion rate differs between three traffic sources.

Part F Experiments

13. A product manager compares customers who opted into a new onboarding flow with those who did not, finds the opted-in group retains better, and proposes rolling it out.
- (a) (3 points) Name what randomization would have bought that this comparison cannot.

 - (b) (2 points) Name one threat that randomization alone does **not** remove.

*A correct number with no sentence around it earns part marks.
Every part that asks what a number means is graded on the sentence.*